

Authoring framework for industrial XR digital twin and autonomous agent: a proof of concept

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ABSTRACT

Extended Reality-Interfaced Digital Twins (XR-DTs) can be used in a wide range of industrial applications, including training, simulation, and the design and monitoring of industrial systems. However, interacting with and controlling autonomous agents, whether virtual or real, through XR-DT can be challenging, and existing solutions often address only a certain level of collaboration or limited use cases. To address these limitations, this paper presents a no-code scenario authoring solution for XR-DTs called NEURONES as well as a demonstration scenario.

Index Terms: Industry 5.0, Extended Reality, Digital Twin, Scenario authoring, Human-Robot collaboration, Autonomous agents

1 INTRODUCTION

Industry 5.0 was built upon the technological foundations from Industry 4.0, such as the Internet of Things, Augmented Reality (AR), Virtual Reality (VR), Artificial Intelligence and Digital Twins (DT). The latter are digital replicas of a physical environment which can communicate bi-directionally with their Physical Twin (PT) to send updates on their states. They can be merged with eXtended Reality (XR), which combine immersive technologies from AR to VR, to allow users to interact with both real and virtual systems from both real and virtual worlds [2]. XR-DTs can be used in a wide range of industrial applications, including monitoring, maintenance, training, production system design, remote robot control, production activities or pre-production simulation.

One of the challenges associated with XR-DTs is the scarcity of tools available for scenario authoring within these environments [3]. Scenarios, which define various tasks to be performed, represent a type of content that would be valuable to the industry as they can address a wide range of use cases. Depending on the situation, a scenario can, for example, guide an operator with AR instructions during a maintenance or train him in a VR simulation of assembly procedures assisted by a robot. Another example is the design of the production system involving the creation and the evaluation of scenarios with XR-DT simulation prior to deployment on the PT.

The purpose of this paper is to introduce NEURONES, a generic framework for authoring scenarios within XR-DT environments and an example of integration inside a DT architecture thanks to a demonstration scenario wherein a humanoid robot (a Tiago++ robot) collaborates with a human operator in XR.

2 METHODOLOGY

The first step was to identify the functional requirements on the basis of the several identified use cases, including monitoring, maintenance, production activities, training and production system design as well as on an in-depth literature review. These requirements are

Table 1: XR-DT authoring frameworks functional requirements matrix based on use case

	Autonomous agents	Data transfer between twins	Deployment on the PT
Monitoring	X	X	
Maintenance/ Production	X	X	
Training	X		
Production system design	X	X	X
Remote agent control	X	X	X

the involvement of human and non-human agents, the connection and data transfer between the PT and the XR-DT, as well as the possibility of deploying simulated scenarios in the XR-DT to the PT. Table 1 synthesizes the requirements according to the identified use cases. As business experts often lack programming skills, no-code authoring mode through either a graph formalism or immersive methods are the most appropriate. Immersive methods include ‘Authoring-by-doing’ (AbDo) [5] which allows actions from the users to be transformed into steps of the scenario as they are completed, or ‘Authoring-by-demonstration’ (AbDe) [1] where the user manipulates the robot to show it how to perform the required action.

Using these criteria, the INTERVALES [5] framework was chosen as a basis of the solution. It follows an entity-feature paradigm in which objects users can interact with are called **entities** and are composed of **features** that define the behavior of the entity. Interacting in XR with an entity produces **actions** which can be translated into Tasks when using AbDo, allowing to quickly create and iterate on scenarios. It is coupled with a node editor in order to create non-sequential scenarios containing parallelization and branching. Despite lacking functionalities related to the notion of agent, the modularity of the framework makes it easy to add the missing functionalities.

The NEURONES model is a major evolution of the INTERVALES model, with the primary objective of expanding the authoring capabilities of the framework by integrating agents within scenarios. The main changes Table 2 are a more advanced implementation of agents, which are entities capable of interacting with their environment by performing actions according to their **competences**. These actions are carried out by at least one agent, which can be either a human equipped with any XR device, or an autonomous agent, such as, for example, a virtual or robotic agent. Autonomous agents can be integrated in the scenario by using AbDe.

Table 2: Comparison of functionalities between INTERVALES and NEURONES

Functional requirement	INTERVALES	NEURONES
Autonomous agents in scenario	Agent must be hard-coded	Tasks are dispatched and executed by agents
Connection to the digital twin	Absent	Support OPC-UA or ROS based DT architectures, pluggable to proprietary API
Deployment on the PT	Unavailable	Can dispatch tasks of a scenario to the real agents

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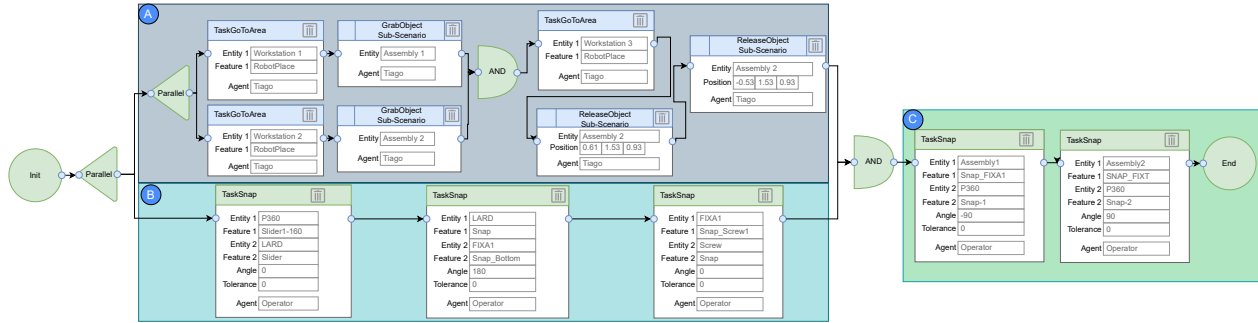


Figure 1: The demonstration scenario. The colors represent the type of agent the task is given to: blue tasks are attributed to the autonomous agent, green tasks to the human agent.

3 RESULTS

To ensure the applicability of the presented application, it has been integrated in a first proof of concept. It is based on a flexible manufacturing assembly process in which a dual-arm Tiago++ humanoid robot (Tiago) collaborates with a human operator on an assembly task described in the scenario represented in Figure 1. A **Parallel node** separates the tasks of the operator assembling parts (Fig. 1-A) from those of the Tiago delivering assemblies (Fig. 1-C). Once both agents have completed their tasks, the operator combines all the assemblies (Fig. 1-C).

The XR-DT architecture follows the methodology presented in [4]. It uses the OPC-UA communication protocol as the backbone of the XR-DT (Fig. 2). On the left side, the PT (Fig. 2-A) contains all the dynamic elements connected to the twin: the Tiago and the workstations. The OPC-UA servers (Fig. 2-B) act as interoperable interfaces, both receiving requests from NEURONES's agent competences to execute tasks and sending data updates. The servers thus manage either the non-intelligent elements of the workshop, like the workstations, or the Tiago by using ROS topics. On the right side, the XR-DT (Fig. 2-C), is divided into two parts. First, The simulation section (Fig. 2-D) consists of docker containers powering a SymPy python server for production flow simulation, a ROS simulation for the Tiago and its OPC-UA server. Secondly, the XR application integrating NEURONES (Fig. 2-E) is the entry point of the XR-DT.

This proof of concept thus illustrates how NEURONES, when integrated within a DT architecture, enables the creation of non-linear scenarios usable for different use cases in an XR-DT, as the demonstration scenario could be used in a training situation where a user learns the collaborative assembly processes in VR or in an XR-DT simulation to both set up and validate a new workshop layout design prior to its deployment on the PT for robot tasks orches-

tration and operator support in manufacturing operations thanks to AR.

4 CONCLUSION

In the context of Industry 5.0 and XR-DT, this paper presents the NEURONES framework and explores how it is used to build a bridge between both environments, physical and digital, and the scenarization models proposed in INTERVALES. As a result, industrial experts lacking programming skills or knowledge can interact with agents such as robotic agents in the XR-DT to author and edit scenarios involving them and to deploy to their real counterparts. The NEURONES framework was built upon INTERVALES after a comparative analysis of several tools and frameworks, with a focus on either autonomous agents, connection to XR-DTs and offering a no-code approach to author XR content.

Future work should focus on user evaluation to ensure a hybrid approach between AbDe, AbDo and node modeling allows users with neither robotics nor computing skills to create complex scenarios involving autonomous agents in the XR-DT.

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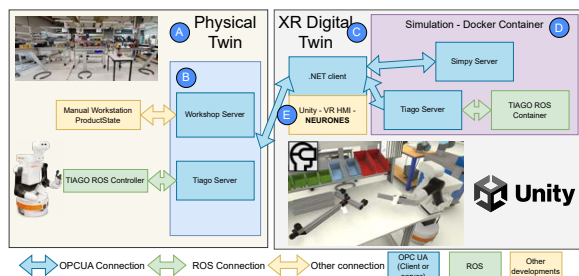


Figure 2: The XR-DT architecture